

IMAT MOCK EXAM 5 — "The Discriminator"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 42/90.0 · Mock exam · answer + worked solution follow every question

Each question is immediately followed by its answer and worked solution, so this paper is for review and paper study rather than blind simulation. Letters refer to the option order printed here; the online app shuffles options on every attempt.

Section A - Logical Reasoning & Reading (Q1-9)

1. "In oxidative phosphorylation, the proton gradient across the inner mitochondrial membrane is coupled to ATP synthesis. If the membrane is made permeable to protons by an uncoupler such as dinitrophenol, the gradient dissipates, electron transport accelerates, but ATP synthesis falls. Heat is produced instead."

Which is the strongest inference?

- A) Uncouplers raise ATP yield.
- B) The proton gradient is independent of electron transport.
- C) Uncoupling disconnects electron transport from ATP synthesis.
- D) Without a proton gradient, ATP synthesis rises.
- E) Uncoupling stops electron transport.

Answer: C

The passage lists three simultaneous effects of the uncoupler: the proton gradient dissipates, electron transport accelerates, and ATP synthesis falls as the energy escapes as heat. The only statement consistent with all three is that uncoupling breaks the coupling between electron transport and ATP synthesis: oxidation runs on, even faster, while phosphorylation stops. The claim that uncoupling stops electron transport is directly contradicted by the stated acceleration, while the claims that ATP synthesis rises without the gradient, that uncouplers raise ATP yield, and that the gradient is independent of electron transport all contradict the fall in ATP output.

2. "A population bottleneck occurs when a population's size is drastically reduced for at least one generation. The genetic diversity of the surviving population is typically much lower than that of the original, because alleles are lost by chance. After the bottleneck, even rare alleles that remain can rise in frequency by genetic drift."

Which follows?

- A) Genetic drift stops after a bottleneck.
- B) Bottlenecks have no evolutionary consequence.
- C) Bottlenecks reduce genetic diversity regardless of which alleles happen to survive.
- D) After a bottleneck, the surviving diversity depends partly on which alleles randomly survived.
- E) Bottlenecks always raise genetic diversity.

Answer: D

The passage states that alleles are lost by chance during the bottleneck and that surviving rare alleles may afterwards rise by drift, so the diversity that persists and evolves depends partly on the random subset of alleles that survived. The claim that bottlenecks always raise diversity contradicts the stated loss of alleles, and saying drift stops afterwards contradicts the passage's explicit role for drift. Even the broader claim that diversity falls regardless of which alleles survive misses the point: the outcome hinges on which alleles chance kept, which is exactly what the correct inference says.

3. (General knowledge) The Italian physician and educator who developed an eponymous educational method emphasising self-directed activity in a prepared environment, and was nominated for the Nobel Peace Prize multiple times, was:

- A) Rita Levi-Montalcini
- B) Maria Montessori
- C) Elena Lucrezia Cornaro Piscopia
- D) Sofia Vanni
- E) Nilde Iotti

Answer: B

Maria Montessori (1870 to 1952) developed the Montessori method, built on self-directed activity within a prepared environment, and was nominated several times for the Nobel Peace Prize. Rita Levi-Montalcini was a Nobel laureate neuroembryologist and Elena Lucrezia Cornaro Piscopia the first woman to receive a university doctorate, so neither matches the educational-method description.

4. Five people — F, G, H, I, J — are in a queue. F is immediately behind G. H is somewhere in front of G. I is not at either end. J is at the back. Which must be true?

- A) G is at the front.
- B) I is between H and G.
- C) F is immediately in front of J.
- D) F is third.
- E) H is at the front.

Answer: E

Givens: five queue spots numbered 1 (front) to 5 (back); F immediately behind G; H somewhere in front of G; I not at either end; J at the back.

Place J first: J takes spot 5.

The adjacent pair G,F must sit F directly behind G; slots (1,2) are impossible because G at the front leaves no room for H, and slots (4,5) are blocked by J, so only (2,3) and (3,4) remain.

Case (G,F) = (3,4): spots 1 and 2 hold H and I; I cannot take end spot 1, so I = 2 and H = 1, giving H, I, G, F, J.

Case (G,F) = (2,3): H must be somewhere in front of G and the only spot ahead of G left is 1, so H = 1 and I = 4, which is not an end, giving H, G, F, I, J.

Result: exactly two orders satisfy all four rules and H is first in both, so the claim that must be true is: H is at the front.

The tempting claim that F is immediately in front of J holds only in the first order; in the second, I sits between F and J, so it is not forced.

5. "All metals conduct electricity. X conducts electricity." From these, we can conclude:

- A) X might or might not be a metal (affirming the consequent is invalid).
- B) X is a metal.
- C) No metal is X.
- D) X is not a metal.
- E) X is an insulator.

Answer: A

Givens: (1) all metals conduct electricity, i.e. metal $\rightarrow$ conducts; (2) X conducts electricity.

The rule runs one way only: from metal you may infer conducts, but nothing given lets you run it backwards.

Walk: X conducts, so X is either a metal or one of the many non-metal conductors such as graphite or electrolyte solutions; both possibilities stay open.

Deduction: X might or might not be a metal.

The tempting claim that X is a metal commits the fallacy of affirming the consequent: from $P \rightarrow Q$ and Q , P does not follow.

6. A number, when divided by 3, gives remainder 2; when divided by 5, gives remainder 3. The smallest positive such number is:

- A) 5 B) 7 C) 8 D) 11 E) 13

Answer: C

Given: n divided by 3 leaves remainder 2; n divided by 5 leaves remainder 3; find the smallest positive n .

Relationship: any candidate must equal $5k + 3$ ($k = 0, 1, 2, \dots$), so test the list 3, 8, 13, ... against the divide-by-3 condition.

Test $n = 3$: $3 = 3 \times 1 + 0$, remainder 0, not 2, so 3 fails

Test $n = 8$: $8 = 3 \times 2 + 2$, remainder 2 (condition met), and $8 = 5 \times 1 + 3$, remainder 3 (condition met).

Result: the smallest positive number is 8.

The tempting 5 does leave remainder 2 on division by 3, but $5 = 5 \times 1 + 0$ leaves remainder 0 on division by 5, not 3.

7. A clock reads 8:20. What is the angle between the hands?

- A) 0° B) 100° C) 120° D) 130° E) 140°

Answer: D

Given: clock time 8:20; the minute hand sweeps 6° per minute and the hour hand 30° per hour plus 0.5° per minute, both measured clockwise from 12.

Relationship: minute hand angle = minutes $\times 6^\circ$; hour hand angle = hours $\times 30^\circ$ + minutes $\times 0.5^\circ$; the answer is their difference.

Substitute: minute hand = $20 \times 6^\circ = 120^\circ$

Hour hand = $8 \times 30^\circ + 20 \times 0.5^\circ = 240^\circ + 10^\circ = 250^\circ$

Difference = $250^\circ - 120^\circ = 130^\circ$

Result: the angle between the hands is 130° .

The tempting 120° places the hour hand exactly on the 8 and forgets it creeps 10° past during those 20 minutes.

8. If the 5th of a month is a Tuesday, what day is the 27th of the same month?

- A) Tuesday B) Wednesday C) Thursday D) Saturday E) Friday

Answer: B

Given: the 5th of the month is a Tuesday; find the weekday of the 27th.

Relationship: weekdays repeat every 7 days, so only the gap between the dates matters.

$$\text{Gap} = 27 - 5 = 22 \text{ days}$$

Split the gap: $22 = 7 \times 3 + 1$, i.e

three full weeks plus 1 leftover day.

Shift: Tuesday + 3 weeks = Tuesday; + 1 day = Wednesday.

Result: the 27th falls on a Wednesday.

The tempting Tuesday forgets the leftover 1 day and treats 22 days as exactly three weeks.

9. Complete the letter series: A, C, F, J, O, ?

- A) S B) T C) V D) W E) U

Answer: E

Given: the letter series A, C, F, J, O, ?.

Relationship: convert letters to alphabet positions and inspect the gaps.

Positions: A = 1, C = 3, F = 6, J = 10, O = 15

Gaps: $3 - 1 = 2$, $6 - 3 = 3$, $10 - 6 = 4$, $15 - 10 = 5$, so the steps are +2, +3, +4, +5 and the next step is +6.

Next term = $15 + 6 = 21$, and the 21st letter is U

Result: U.

The tempting T merely repeats the last gap of 5; the gaps themselves grow by 1 each step.

Section B - Biology (Q10-32)

10. In glycolysis, the committed (rate-limiting) step is catalysed by:

- A) Phosphofructokinase-1 (PFK-1)
B) Aldolase
C) Phosphoglycerate kinase
D) Hexokinase
E) Pyruvate kinase

Answer: A

The committed, rate-limiting step of glycolysis is the phosphorylation of fructose-6-phosphate to fructose-1,6-bisphosphate by phosphofructokinase-1, the pathway's principal allosteric control point (inhibited by ATP, activated by AMP). Hexokinase acts earlier and pyruvate kinase last, but neither is the committed control point: once PFK-1 has acted, the sugar cannot easily leave the glycolytic route.

11. In the electron transport chain, FADH₂ donates electrons to:

- A)** Complex II (succinate dehydrogenase)
- B)** Complex IV
- C)** ATP synthase
- D)** Complex III
- E)** Complex I

Answer: A

FADH₂ is reoxidised by succinate dehydrogenase, which is itself Complex II of the chain, so its electrons enter downstream of Complex I and fewer protons are pumped (about 1.5 ATP per FADH₂ versus about 2.5 per NADH). Assigning FADH₂ to Complex I describes NADH, and ATP synthase is not an electron carrier at all.

12. The chemiosmotic synthesis of ATP in mitochondria is driven by the flow of protons through ATP synthase from:

- A)** Cytosol to matrix
- B)** Intermembrane space to matrix
- C)** Matrix to intermembrane space
- D)** Outer to inner membrane (no gradient)
- E)** Matrix to cytosol

Answer: B

The respiratory chain pumps protons out of the matrix into the intermembrane space, storing energy as an electrochemical gradient. ATP synthase then allows protons to flow back down that gradient, from intermembrane space to matrix, and this return flow powers ATP synthesis. The reverse direction (matrix to intermembrane space) is the pumping step, not the flow through ATP synthase.

13. Under anaerobic conditions, yeast cells regenerate NAD⁺ by reducing pyruvate to:

- A)** Water
- B)** Acetyl-CoA
- C)** Ethanol and CO₂
- D)** Glucose
- E)** Lactate

Answer: C

Yeast performs alcoholic fermentation: pyruvate is decarboxylated to acetaldehyde (releasing CO₂), and acetaldehyde is reduced to ethanol, oxidising NADH back to NAD⁺ so glycolysis can continue. Lactate is the product in oxygen-starved mammalian muscle and in lactic acid bacteria, not in yeast.

14. During muscle contraction, calcium ions released from the sarcoplasmic reticulum bind to:

- A)** Myosin heads
- B)** Actin directly
- C)** The sarcolemma only
- D)** ATP
- E)** Troponin, causing tropomyosin to move and expose myosin-binding sites

Answer: E

Ca²⁺ released from the sarcoplasmic reticulum binds troponin C; the troponin complex then shifts tropomyosin away from actin's myosin-binding sites, permitting cross-bridge formation. Calcium does not trigger contraction by binding the myosin heads or actin directly.

15. The sliding filament theory states that during contraction:
- A) The Z-lines move apart.
 - B) Actin filaments shorten.
 - C) Myosin filaments shorten.
 - D) The sarcomere shortens, but the filaments themselves do not change length (they slide past each other).
 - E) Both filaments shorten in proportion.

Answer: D

In the sliding filament model the thick and thin filaments keep constant length and simply slide past one another, pulling the Z-lines toward the sarcomere centre; the sarcomere shortens while neither filament does. Each claim that actin or myosin (or both) shortens contradicts the model's central point, and the Z-lines moving apart is the opposite of what happens.

16. Slow-twitch (Type I) muscle fibres are specialised for:
- A) Replacing damaged neurons
 - B) Generating action potentials
 - C) Storing calcium
 - D) Rapid, powerful, short-duration contractions using anaerobic glycolysis
 - E) Endurance activity using oxidative metabolism (many mitochondria, high myoglobin)

Answer: E

Type I (slow-twitch, red) fibres are specialised for endurance: abundant mitochondria, high myoglobin, dense capillaries and predominantly oxidative metabolism make them fatigue-resistant. The description of rapid, powerful, short-duration anaerobic bursts matches fast-twitch (Type II) fibres instead.

17. Most CO₂ transported in the blood is carried:
- A) Dissolved as CO₂ gas in plasma
 - B) As bicarbonate ions (HCO₃⁻) in plasma
 - C) Bound to albumin
 - D) Within erythrocyte nuclei
 - E) Bound to haemoglobin (carbaminohaemoglobin)

Answer: B

About 70% of blood CO₂ travels as bicarbonate: CO₂ enters red blood cells, carbonic anhydrase converts it to carbonic acid, which dissociates to H⁺ and HCO₃⁻, and the bicarbonate is exported into the plasma. Only about 23% rides on haemoglobin as carbaminohaemoglobin and about 7% dissolves in plasma, so neither of those is the majority route.

18. The enzyme carbonic anhydrase, abundant in red blood cells, catalyses:
- A) $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$
 - B) Glycolysis
 - C) The conversion of haemoglobin to oxyhaemoglobin
 - D) Pyruvate → acetyl-CoA
 - E) The clotting cascade

Answer: A

Carbonic anhydrase catalyses the reversible hydration $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}_2\text{CO}_3$, which then dissociates into H⁺ and HCO₃⁻; this reaction underpins CO₂ transport and the red cell's buffering of blood pH. It has no role in haemoglobin oxygenation, glycolysis, clotting or the pyruvate to acetyl-CoA step.

19. In the nephron, the majority of filtered water and solutes is reabsorbed in the:

- A) Loop of Henle (descending limb only)
- B) Distal convoluted tubule
- C) Collecting duct
- D) Proximal convoluted tubule (~65%)
- E) Renal pelvis

Answer: D

The proximal convoluted tubule reabsorbs roughly 65% of the filtered Na^+ , water, glucose and amino acids, by far the largest share of any nephron segment. The distal tubule and collecting duct only fine-tune what remains, and the renal pelvis merely drains urine.

20. Antidiuretic hormone (ADH) increases water reabsorption by inserting aquaporins into the:

- A) Afferent arteriole
- B) Proximal tubule
- C) Collecting duct (principal cells)
- D) Glomerulus
- E) Loop of Henle (thin limb)

Answer: C

ADH binds V2 receptors on collecting duct principal cells and triggers insertion of aquaporin-2 channels into the apical membrane, sharply increasing the duct's water permeability and hence reabsorption. The glomerulus filters rather than reabsorbs, and proximal water reabsorption follows solutes osmotically without ADH control.

21. Aldosterone, secreted by the adrenal cortex, acts on the kidney to increase reabsorption of:

- A) Glucose
- B) Urea
- C) K^+
- D) Albumin
- E) Na^+ (and water), and increase K^+/H^+ excretion

Answer: E

Aldosterone from the adrenal cortex acts on the distal tubule and collecting duct to increase Na^+ reabsorption (water follows passively) while promoting K^+ and H^+ secretion. The option asserting increased K^+ reabsorption states the opposite of the hormone's actual effect on potassium.

22. In nervous tissue, the myelin sheath is produced in the CNS by __ and in the PNS by __.

- A) Oligodendrocytes / Schwann cells
- B) Ependymal cells / astrocytes
- C) Astrocytes / Schwann cells
- D) Microglia / satellite cells
- E) Neurons / oligodendrocytes

Answer: A

In the CNS, oligodendrocytes wrap myelin around axon segments (one cell serving several axons); in the PNS, Schwann cells myelinate single segments. The pairing astrocytes with Schwann cells fails because astrocytes are supporting and blood-brain barrier cells, not myelin formers, and microglia are the CNS's phagocytes.

23. During an action potential, the rapid depolarisation is due to:

- A)** K^+ efflux
- B)** Ca^{2+} influx at the soma
- C)** Na^+ influx (voltage-gated Na^+ channels open)
- D)** Cl^- influx
- E)** Active transport of Na^+ out

Answer: C

The rapid upstroke of the action potential is produced by the opening of voltage-gated Na^+ channels: Na^+ floods in down its electrochemical gradient and drives the potential toward about +30 mV. K^+ efflux through delayed channels causes the repolarising falling phase, and active Na^+ extrusion by the Na^+/K^+ pump is thousands of times too slow to account for a millisecond-scale depolarisation.

24. A cross between two plants both with red flowers produces 3 red : 1 white offspring. This is consistent with:

- A)** Polygenic inheritance
- B)** Complete dominance (red dominant, parents heterozygous)
- C)** Sex-linkage
- D)** Incomplete dominance
- E)** Codominance

Answer: B

A 3 red : 1 white ratio from two red parents is the classic monohybrid result of two heterozygotes under complete dominance: the 1:2:1 genotypic ratio collapses to 3 dominant phenotypes per 1 recessive. Incomplete dominance would instead yield a 1:2:1 phenotype ratio with an intermediate class, and codominance would express both traits side by side; neither produces 3:1.

25. A man with blood group A (genotype AO) and a woman with blood group B (genotype BO) can have children with which blood groups?

- A)** O only
- B)** A only
- C)** B only
- D)** A, B, AB, and O
- E)** AB only

Answer: D

The father (AO) passes on A or O and the mother (BO) passes on B or O, so the offspring genotypes are AB, AO (group A), BO (group B) and OO (group O). All four ABO phenotypes are therefore possible. Any answer restricting the children to a single group discards three of the four equally likely combinations.

26. DNA polymerase synthesises new DNA in which direction?

- A)** $3' \rightarrow 5'$ on the new strand
- B)** From the C-terminus
- C)** Without direction
- D)** In both directions simultaneously on one strand
- E)** $5' \rightarrow 3'$ on the new strand

Answer: E

DNA polymerase can only attach a nucleotide to a free $3'$ -OH end, so every new strand grows in the $5'$ to $3'$ direction while its template is read $3'$ to $5'$. Growth in the $3'$ to $5'$ direction on the new strand is chemically impossible, which is exactly why the lagging strand is built discontinuously as Okazaki fragments.

27. The genetic code is described as "degenerate" because:

- A) Codons overlap.
- B) One codon codes for multiple amino acids.
- C) The code is ambiguous (one codon → multiple amino acids).
- D) Multiple codons can code for the same amino acid.
- E) The code differs between species.

Answer: D

The code is degenerate because most amino acids are specified by more than one codon (leucine by six), a consequence of 61 sense codons coding for about 20 amino acids. Degeneracy is not ambiguity: each individual codon still specifies exactly one amino acid, and the code is essentially universal across species.

28. A bacterial operon whose transcription is increased in the presence of a small molecule (e.g., lactose for the lac operon) is:

- A) Inducible (the molecule is an inducer)
- B) Constitutive
- C) Repressible
- D) Always off
- E) Negatively regulated only

Answer: A

An operon whose transcription increases when a small molecule is present is inducible, and that molecule is the inducer; for the lac operon, allolactose inactivates the lac repressor and transcription rises. A repressible operon behaves oppositely (transcription falls when the co-repressor, e.g. tryptophan for the trp operon, is present), and constitutive means expressed regardless of the molecule.

29. Which cell type produces antibodies?

- A) Macrophages
- B) Platelets
- C) B cells (plasma cells, the differentiated form)
- D) Mast cells
- E) T-helper cells

Answer: C

Antibodies are secreted by plasma cells, the terminally differentiated form of B lymphocytes. T-helper cells coordinate the response via cytokines, macrophages phagocytose, and mast cells release histamine, but none of them secretes antibody.

30. In photosynthesis, the oxygen released comes from:

- A) O₂ taken from the air
- B) H₂O (water, split in the light reactions)
- C) Glucose
- D) CO₂
- E) NADPH

Answer: B

The O₂ released comes from splitting water at photosystem II's oxygen-evolving complex: $2 \text{H}_2\text{O} \rightarrow 4 \text{H}^+ + 4 \text{e}^- + \text{O}_2$. Isotope labelling settled it: water labelled with ¹⁸O yields labelled O₂, whereas labelled CO₂ does not, ruling out carbon dioxide or glucose as the oxygen source.

31. In a DNA double helix, 20% of bases are adenine. The percentage of guanine is:

- A) 20% B) 30% C) 40% D) 60% E) 80%

Answer: B

By Chargaff's rules $A = T$ and $G = C$ in double-stranded DNA. With $A = 20\%$, T is also 20% , leaving $100 - 40 = 60\%$ to split equally between G and C , so $G = 30\%$. The answer of 20% would wrongly make all four bases equal, and 60% is the combined $G + C$ fraction, not G alone.

32. A mutation that inserts a single nucleotide into a coding sequence (not in multiples of 3) is called a:

- A) Point mutation only
B) Nonsense mutation
C) Silent mutation
D) Missense mutation
E) Frameshift mutation

Answer: E

Inserting one nucleotide, since 1 is not a multiple of 3, shifts the triplet reading frame for every codon downstream, so this is a frameshift mutation. Missense and nonsense mutations change one codon at a time without disturbing the frame, and silent mutations change no amino acid at all.

Section C - Chemistry (Q33-47)

33. How many grams of CO_2 are produced when 0.5 mol of C reacts completely with O_2 ? ($\text{C} = 12$, $\text{O} = 16$.)

- A) 4.4 g B) 11 g C) 22 g D) 44 g E) 88 g

Answer: C

Given: 0.5 mol of carbon burned completely in O_2 ; $M(\text{C}) = 12 \text{ g/mol}$, $M(\text{O}) = 16 \text{ g/mol}$

Relationship: $\text{C} + \text{O}_2 \rightarrow \text{CO}_2$ shows a 1:1 mole ratio of C to CO_2 ; mass = moles $\times$ molar mass.

Substitute: moles of $\text{CO}_2 = 0.5 \text{ mol} \times 1 = 0.5 \text{ mol}$

$M(\text{CO}_2) = 12 + (2 \times 16) = 44 \text{ g/mol}$

Mass = $0.5 \text{ mol} \times 44 \text{ g/mol} = 22 \text{ g}$

Result: 22 g of CO_2 .

The tempting 11 g corresponds to only 0.25 mol, as if the 1:1 ratio halved the amount.

34. Balance: $__ \text{Fe} + __ \text{O}_2 \rightarrow __ \text{Fe}_2\text{O}_3$. The coefficient of Fe is:

- A) 1 B) 2 C) 3 D) 4 E) 6

Answer: D

Given: the skeleton equation $\text{Fe} + \text{O}_2 \rightarrow \text{Fe}_2\text{O}_3$.

Relationship: balance by atom counting; each Fe_2O_3 unit contains 2 Fe atoms and 3 O atoms.

Take 2 Fe_2O_3 : that fixes 4 Fe atoms and $2 \times 3 = 6$ O atoms on the right.

Supply Fe: 4 Fe on the left.

Supply O: 6 O atoms require 3 O_2 molecules.

Check: $\text{Fe } 4 = 4$; $\text{O } 3 \times 2 = 6 = 2 \times 3$

Result: $4 \text{ Fe} + 3 \text{ O}_2 \rightarrow 2 \text{ Fe}_2\text{O}_3$, so the coefficient of Fe is 4.

The tempting 3 copies the 3 from the oxygen side without re-checking that the iron atoms then fail to balance.

35. What volume of 0.1 M H_2SO_4 is needed to neutralise 25 mL of 0.2 M NaOH? (H_2SO_4 is diprotic.)

- A) 5 mL B) 12.5 mL C) 25 mL D) 50 mL E) 100 mL

Answer: C

Given: 25 mL of 0.2 M NaOH; H_2SO_4 at 0.1 M, diprotic (2 H^+ per molecule).

Relationship: neutralisation needs equal moles of H^+ and OH^- ; moles = volume $\times$ molarity.

Substitute: moles NaOH = $0.025 \text{ L} \times 0.2 \text{ mol/L} = 0.005 \text{ mol}$, i.e. 0.005 mol OH^- to neutralise

Moles H_2SO_4 = $0.005 \text{ mol} \div 2 = 0.0025 \text{ mol}$

Volume = $0.0025 \text{ mol} \div 0.1 \text{ mol/L} = 0.025 \text{ L} = 25 \text{ mL}$

Result: 25 mL of the acid.

The tempting 12.5 mL treats sulfuric acid as monoprotic and skips the division by 2.

36. The pH of a buffer containing 0.1 M acetic acid ($\text{pK}_a = 4.76$) and 0.1 M sodium acetate is:

- A) 3.76 B) 4.76 C) 5.76 D) 7.0 E) 9.24

Answer: B

Given: [acetic acid] = 0.1 M, [acetate] = 0.1 M, $\text{pK}_a = 4.76$.

Relationship: Henderson-Hasselbalch, $\text{pH} = \text{pK}_a + \log([\text{A}^-]/[\text{HA}])$.

Substitute: ratio = $0.1 \div 0.1 = 1$

$\log 1 = 0$, so $\text{pH} = 4.76 + 0 = 4.76$

Result: pH 4.76; a buffer with equal acid and conjugate base sits exactly at its pK_a .

The tempting 3.76 or 5.76 assumes a tenfold ratio between conjugate base and acid; here the ratio is exactly 1.

37. The oxidation number of oxygen in hydrogen peroxide (H_2O_2) is:

- A) -2 B) -1 C) 0 D) $+1$ E) $+2$

Answer: B

Given: H_2O_2 , a neutral molecule; hydrogen counts as $+1$ when bonded to non-metals.

Relationship: oxidation numbers sum to 0 in a neutral compound.

Substitute: $2 \times (+1) + 2x = 0$

Solve: $2 + 2x = 0$, so $2x = -2$, $x = -1$

Result: oxygen is -1 in H_2O_2 , the peroxide exception.

The tempting -2 is the default oxide value, but it fails the check: $2 \times (+1) + 2 \times (-2) = -2$, not 0.

38. Which species is reduced in: $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$?

- A) Cl_2 ($\text{Cl}_2 \rightarrow 2\text{Cl}^-$)
- B) Na ($\text{Na} \rightarrow \text{Na}^+$)
- C) Nothing
- D) NaCl
- E) Both Na and Cl_2

Answer: A

Given: $2\text{Na} + \text{Cl}_2 \rightarrow 2\text{NaCl}$; uncombined elements have oxidation number 0.

Relationship: reduction is gain of electrons, seen as a fall in oxidation number; oxidation is a rise.

Before: Na = 0, Cl = 0 (free elements)

After: in NaCl, Na = +1 (Group 1 metal), so Cl = -1 because the compound is neutral.

Track: Na goes $0 \rightarrow +1$, losing 1e^- per atom (oxidised); Cl goes $0 \rightarrow -1$, gaining 1e^- per atom (reduced).

Result: Cl_2 is the species reduced.

The tempting claim that both species are reduced cannot hold: the electrons sodium loses are exactly the electrons chlorine gains, one loss for every gain.

39. An ideal gas occupies 2 L at 1 atm and 300 K. What is its volume at 2 atm and 300 K?

- A) 0.5 L B) 1 L C) 2 L D) 4 L E) 8 L

Answer: B

Given: $V_1 = 2\text{ L}$ at $P_1 = 1\text{ atm}$, then $P_2 = 2\text{ atm}$ at constant $T = 300\text{ K}$.

Relationship: Boyle's law, $P_1V_1 = P_2V_2$ (temperature and amount fixed).

Substitute: $1\text{ atm} \times 2\text{ L} = 2\text{ atm} \times V_2$

Rearrange: $V_2 = (1 \times 2) \div 2 = 1\text{ L}$

Result: 1 L; doubling the pressure halves the volume.

The tempting 4 L reads pressure and volume as directly proportional instead of inversely.

40. How many electrons can occupy the d-subshell (5 orbitals)?

- A) 2 B) 6 C) 8 D) 10 E) 14

Answer: D

Given: a d subshell contains 5 orbitals; each orbital holds at most 2 electrons of opposite spin.

Relationship: subshell capacity = number of orbitals $\times$ 2 electrons.

Substitute: $5 \times 2 = 10\text{ electrons}$

Result: 10 electrons.

The tempting 6 or 14 borrows the orbital counts of the p subshell (3 orbitals) or the f subshell (7 orbitals).

41. Which element has the electron configuration $1s^2 2s^2 2p^6 3s^1$?

- A) Na B) K C) Mg D) Al E) Ne

Answer: A

Given: the electron configuration $1s^2 2s^2 2p^6 3s^1$.

Relationship: the superscripts count electrons; for a neutral atom that equals the atomic number.

Substitute: $2 + 2 + 6 + 1 = 11$ electrons

Atomic number 11 is sodium.

Result: Na.

The tempting Mg would need 12 electrons ($3s^2$); the given configuration ends in a single 3s electron.

42. The geometry of NH_3 (ammonia) is:

- A) Trigonal planar
B) Octahedral
C) Square planar
D) Linear
E) Tetrahedral (electron-pair geometry) / trigonal pyramidal (molecular shape)

Answer: E

Nitrogen in NH_3 has three bonding pairs plus one lone pair, four electron domains in all, so the electron-pair geometry is tetrahedral; the lone pair occupies one corner, giving the molecular shape of a trigonal pyramid with a bond angle near 107° . Trigonal planar would require the lone pair's absence, and linear or octahedral arrangements fit neither the domain count nor the observed shape.

43. Which compound is most soluble in water?

- A) CH_3OH (methanol)
B) CCl_4
C) CH_4 (methane)
D) C_6H_6 (benzene)
E) C_2H_6 (ethane)

Answer: A

Methanol carries a polar O-H group and can donate and accept hydrogen bonds with water, making it miscible with water in all proportions. Methane, ethane, benzene and carbon tetrachloride are essentially non-polar and cannot hydrogen-bond with water, so all dissolve far less.

44. An exothermic reaction has:

- A) No activation energy
B) Products higher in energy than reactants
C) $\Delta H > 0$
D) $\Delta H = 0$
E) $\Delta H < 0$

Answer: E

An exothermic reaction releases heat to the surroundings, so the products end up lower in enthalpy than the reactants and ΔH is negative. $\Delta H > 0$ is the definition of an endothermic reaction, and every chemical reaction still faces an activation energy barrier regardless of the sign of ΔH .

45. The IUPAC name of $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ is:

- A) Propane B) Ethane C) Methane D) Butane E) Pentane

Answer: D

The chain $\text{CH}_3\text{-CH}_2\text{-CH}_2\text{-CH}_3$ contains four carbon atoms joined only by single bonds, so the alkane is butane, C_4H_{10} . Propane has three carbons and pentane five.

46. For the equilibrium $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$, $\Delta H < 0$. Increasing pressure at constant T will:

- A) Shift left
B) Shift right
C) Have no effect (equal moles of gas on both sides)
D) Decrease the rate
E) Increase ΔH

Answer: C

Le Chatelier's response to pressure only occurs when the change alters the total moles of gas. Both sides of $\text{H}_2(\text{g}) + \text{I}_2(\text{g}) \rightleftharpoons 2\text{HI}(\text{g})$ hold exactly 2 moles of gas, so raising pressure at constant temperature shifts the position of equilibrium not at all (though the rate does increase). Predicting a right or left shift misapplies the principle to a balanced-mole system, and ΔH is a fixed property of the reaction, not of the pressure.

47. If the rate constant doubles when temperature increases by 10°C , and the rate at T_1 is r , the rate at $T_1 + 20^\circ\text{C}$ is approximately:

- A) $2r$ B) $3r$ C) $4r$ D) $8r$ E) $10r$

Answer: C

Given: the rate constant doubles for every 10°C rise; the rate at T_1 is r .

Relationship: consecutive 10°C steps compound, each multiplying the rate by 2.

Step 1: at $T_1 + 10^\circ\text{C}$ the rate is $2 \times r = 2r$

Step 2: at $T_1 + 20^\circ\text{C}$ the rate is $2 \times 2r = 4r$

Result: $4r$.

The tempting $2r$ treats the whole 20°C rise as a single doubling step.

Section D - Mathematics (Q48-54)

48. Solve: $x^2 - 5x - 6 = 0$.

- A) No real roots B) $x = 1, -6$ C) $x = 2, 3$ D) $x = 5, 1$ E) $x = -1, 6$

Answer: E

Given: $x^2 - 5x - 6 = 0$

Relationship: factor as $(x + p)(x + q) = 0$ with $p \times q = -6$ and $p + q = -5$

Find the pair: $-6 \times 1 = -6$ and $-6 + 1 = -5$, so $p = -6$, $q = 1$

Substitute: $(x - 6)(x + 1) = 0$

Zero product rule: $x - 6 = 0$ gives $x = 6$; $x + 1 = 0$ gives $x = -1$.

Result: $x = 6$ or $x = -1$

The tempting pair $+6$ and -1 has the right product but flips the sum to $+5$, which is the standard factorisation slip here.

49. Solve: $\log_3(x) = -2$.

- A) $x = 1/9$ B) $x = 9$ C) $x = -2$ D) $x = -6$ E) $x = 1/3$

Answer: A

Given: $\log_3(x) = -2$, which requires $x > 0$

Relationship: $\log_b(y) = c$ means exactly $y = b^c$.

Substitute: $x = 3^{-2}$

Evaluate: $3^2 = 9$, and the negative exponent takes the reciprocal, so $3^{-2} = 1/9$.

Result: $x = 1/9$

The tempting 9 silently reads the exponent as +2 and drops the reciprocal.

50. Solve: $(2x - 1)/(x + 1) = 3$ ($x \neq -1$).

- A) $x = 1$ B) $x = -2$ C) $x = 2$ D) $x = -4$ E) $x = 4$

Answer: D

Given: $(2x - 1)/(x + 1) = 3$ with $x \neq -1$

Relationship: multiply both sides by $(x + 1)$, legitimate because $x \neq -1$.

Substitute: $2x - 1 = 3(x + 1)$

Expand: $2x - 1 = 3x + 3$

Collect: $2x - 3x = 3 + 1$, so $-x = 4$ and $x = -4$

Check: $(2 \times (-4) - 1) \div ((-4) + 1) = (-9)/(-3) = 3$, as required

Result: $x = -4$

The tempting +4 comes from a sign slip while moving the x terms across.

51. Solve the inequality: $x^2 - 5x + 6 > 0$.

- A) $2 < x < 3$ B) $x > 2$ C) $x < 2$ or $x > 3$ D) $x < 3$ E) All real x

Answer: C

Given: $x^2 - 5x + 6 > 0$.

Relationship: factor, then read the sign of an upward-opening parabola either side of its roots.

Factor: $(x - 2)(x - 3) > 0$, with zeros at $x = 2$ and $x = 3$

Sign walk: below 2 both factors are negative, product positive; between 2 and 3 the signs differ, product negative; above 3 both are positive, product positive.

Result: $x < 2$ or $x > 3$.

The tempting $2 < x < 3$ answers the reversed inequality; there the expression is negative.

52. A circle has circumference 12π . What is its area?

- A) 6π B) 12π C) 24π D) 36π E) 144π

Answer: D

Given: circumference $C = 12\pi$

Relationship: $C = 2\pi r$ gives $r = C \div 2\pi$; area = πr^2

Substitute: $r = 12\pi \div 2\pi = 6$

Area = $\pi \times 6^2 = 36\pi$

Result: 36π .

The tempting 144π squares the diameter (12) instead of the radius; 12π simply reuses the circumference figure as the area.

53. In a right triangle, the side opposite the 30° angle has length 5. What is the length of the hypotenuse?

- A)** 5 **B)** $5\sqrt{2}$ **C)** $5\sqrt{3}$ **D)** 10 **E)** 15

Answer: D

Given: right triangle, side opposite the 30° angle = 5.

Relationship: $\sin 30^\circ = \text{opposite} \div \text{hypotenuse}$, and $\sin 30^\circ = 1/2$.

Substitute: $1/2 = 5 \div \text{hypotenuse}$

Solve: hypotenuse = $5 \times 2 = 10$

Result: the hypotenuse is 10.

The tempting $5\sqrt{3}$ is the longer leg (adjacent to the 30° angle), and $5\sqrt{2}$ belongs to 45-45-90 triangles; neither is the hypotenuse.

54. Two dice are rolled. What is the probability that the sum is 7?

- A)** $1/12$ **B)** $1/9$ **C)** $1/6$ **D)** $5/36$ **E)** $7/36$

Answer: C

Given: two fair dice rolled once each; find $P(\text{sum} = 7)$.

Relationship: $6 \times 6 = 36$ equally likely ordered outcomes; count those totalling 7.

Enumerate: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1), giving 6 favourable outcomes.

Probability = $6 \div 36 = 1/6$

Result: $1/6$.

The tempting $5/36$ is the chance of a total of 6 or 8, which have one favourable pair fewer.

Section E - Physics (Q55-60)

55. A car starts from rest and accelerates uniformly at 3 m/s^2 . How far does it travel in 4 s?

- A)** 12 m **B)** 24 m **C)** 36 m **D)** 48 m **E)** 72 m

Answer: B

Given: $u = 0 \text{ m/s}$ (starts from rest), $a = 3 \text{ m/s}^2$, $t = 4 \text{ s}$

Relationship: $s = ut + \frac{1}{2}at^2$

Substitute: $s = 0 \times 4 + \frac{1}{2} \times 3 \times 4^2$

Compute: $4^2 = 16$, so $s = \frac{1}{2} \times 3 \times 16 = \frac{1}{2} \times 48$

$s = 24 \text{ m}$

Result: 24 m.

The tempting 12 m drops the $\frac{1}{2}$ (using $s = at$); 48 m drops it from the other side by using at^2 directly.

56. A $6\ \Omega$ and a $3\ \Omega$ resistor are connected in parallel across a $12\ \text{V}$ battery. What is the total current from the battery?

- A)** 1 A **B)** 2 A **C)** 3 A **D)** 4 A **E)** 6 A

Answer: E

Given: $R_1 = 6\ \Omega$ and $R_2 = 3\ \Omega$ in parallel across $V = 12\ \text{V}$

Relationship: $1/R = 1/R_1 + 1/R_2$, then $I = V \div R$

Substitute: $1/R = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$, so $R = 2\ \Omega$

Current: $I = 12\ \text{V} \div 2\ \Omega = 6\ \text{A}$

Cross-check by branches: $12 \div 6 = 2\ \text{A}$ and $12 \div 3 = 4\ \text{A}$, and $2\ \text{A} + 4\ \text{A} = 6\ \text{A}$

Result: 6 A from the battery.

The tempting 2 A is only the $6\ \Omega$ branch current, ignoring the second resistor.

57. A $100\ \text{W}$ bulb is operated on a $200\ \text{V}$ supply for $10\ \text{s}$. How much electrical energy is used?

- A)** 20,000 J **B)** 100 J **C)** 200 J **D)** 1000 J **E)** 2000 J

Answer: D

Given: $P = 100\ \text{W}$, $t = 10\ \text{s}$; the $200\ \text{V}$ supply is already reflected in the bulb's $100\ \text{W}$ rating.

Relationship: energy = power $\times$ time, $E = P \times t$.

Substitute: $E = 100\ \text{W} \times 10\ \text{s}$

$E = 1000\ \text{J}$

Result: 1000 J.

The tempting 2000 J multiplies the supply voltage by the time (200×10), as if 200 were the power.

58. A $2\ \text{kg}$ ball moving at $5\ \text{m/s}$ strikes a wall and rebounds at $3\ \text{m/s}$. What is the magnitude of the change in momentum?

- A)** $2\ \text{kg}\cdot\text{m/s}$ **B)** $4\ \text{kg}\cdot\text{m/s}$ **C)** $8\ \text{kg}\cdot\text{m/s}$ **D)** $10\ \text{kg}\cdot\text{m/s}$ **E)** $16\ \text{kg}\cdot\text{m/s}$

Answer: E

Given: $m = 2\ \text{kg}$, incoming speed $5\ \text{m/s}$ toward the wall, rebound speed $3\ \text{m/s}$ in the opposite direction.

Relationship: $\Delta p = m(v_f - v_i)$; take the incoming direction as positive so the rebound velocity carries a minus sign.

Substitute: $v_i = +5\ \text{m/s}$, $v_f = -3\ \text{m/s}$

$\Delta p = 2\ \text{kg} \times (-3 - 5)\ \text{m/s} = 2\ \text{kg} \times (-8\ \text{m/s}) = -16\ \text{kg}\cdot\text{m/s}$

Magnitude = $16\ \text{kg}\cdot\text{m/s}$

Result: $16\ \text{kg}\cdot\text{m/s}$.

The tempting $4\ \text{kg}\cdot\text{m/s}$ subtracts the speeds as plain numbers ($5 - 3 = 2$, times $2\ \text{kg}$) and ignores the reversal of direction.

59. A satellite orbits Earth at a radius where gravitational acceleration is 2 m/s^2 . If Earth's radius is R and the satellite is at $3R$ from the centre, the surface gravitational acceleration is:

- A)** 2 m/s^2 **B)** 6 m/s^2 **C)** 9 m/s^2 **D)** 18 m/s^2 **E)** 27 m/s^2

Answer: D

Given: $g = 2 \text{ m/s}^2$ at orbital radius $r = 3R$; find g at the surface, $r = R$.

Relationship: $g \propto 1/r^2$, so $g_{\text{surface}} = g_{\text{orbit}} \times (r_{\text{orbit}} \div r_{\text{surface}})^2$.

Substitute: $g_{\text{surface}} = 2 \text{ m/s}^2 \times (3R \div R)^2$

Compute: $3^2 = 9$, so $g_{\text{surface}} = 2 \times 9 = 18 \text{ m/s}^2$

Result: 18 m/s^2 .

The tempting 6 m/s^2 scales by $1/r$ instead of $1/r^2$, giving a factor of 3 rather than 9.

60. A 0.5 kg block of iron ($c = 450 \text{ J/kg}\cdot^\circ\text{C}$) is heated from 20°C to 100°C . How much heat is required?

- A)** $9,000 \text{ J}$ **B)** $18,000 \text{ J}$ **C)** $22,500 \text{ J}$ **D)** $45,000 \text{ J}$ **E)** $90,000 \text{ J}$

Answer: B

Given: $m = 0.5 \text{ kg}$, $c = 450 \text{ J/(kg}\cdot^\circ\text{C)}$, $T_i = 20^\circ\text{C}$, $T_f = 100^\circ\text{C}$

Relationship: $Q = m \times c \times \Delta T$ with $\Delta T = T_f - T_i$.

Substitute: $\Delta T = 100^\circ\text{C} - 20^\circ\text{C} = 80^\circ\text{C}$

$Q = 0.5 \text{ kg} \times 450 \text{ J/(kg}\cdot^\circ\text{C)} \times 80^\circ\text{C} = 225 \times 80 \text{ J}$

$Q = 18,000 \text{ J}$

Result: $18,000 \text{ J}$.

The tempting $22,500 \text{ J}$ uses $\Delta T = 100^\circ\text{C}$ and ignores the 20°C starting temperature.